

Leveraging Real-Time EEG Neurofeedback in Virtual Reality towards Personalized Interventions in Exposure Therapy

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ABSTRACT

Innovative therapeutic tools are essential for conditions like anxiety and food aversion. This study integrates EEG and ECG signals with a VR-based serious game for exposure therapy. Preliminary results show consistent detection of EEG biomarkers, such as N200, during VR tasks and evidence of neural habituation. Despite challenges with ECG artifacts, the successful integration of EEG and VR highlights the feasibility of real-time neurofeedback. Our approach shows auspicious results towards personalized therapeutic interventions by combining VR's immersive qualities with the potential of neurofeedback-driven adaptability, advancing exposure therapy based on physiological feedback.

Index Terms: VR—EEG—ECG—Neurofeedback—User Study.

1 INTRODUCTION

Childhood and adolescence are critical for mental and physical development, yet stress and anxiety are common during these periods. Addressing these challenges requires innovative therapeutic approaches. Serious Games (SG), especially when paired with VR, offer immersive 3D environments that address mental health conditions like PTSD, ASD, and depression [3].

Electrophysiological signals, like EEG for brain activity and ECG for cardiac monitoring, add a new dimension to these approaches. Neurofeedback (NF), using real-time physiological signals to regulate brain activity, is a promising therapy solution. EEG-based NF processes signals to provide feedback (e.g., scores or stimuli), fostering neuroplasticity and improving cognitive functions like attention, memory, and relaxation [2]. Key EEG biomarkers, like N200 and P300, have been identified as markers of inhibition and cognitive engagement, making them key to NF applications.

This study integrates EEG, ECG, and VR in a novel system for food aversion therapy, suggesting the feasibility of real-time neurofeedback in immersive environments.

2 USER STUDY - PARTICIPANTS AND PROTOCOL

The study was composed of two stages, preliminary and phase one of the regular user study, which will have a second phase. In both stages, participants were required to complete a survey to ascertain whether any exclusion criteria were applicable. All participants signed an informed consent.

Preliminary stage included six participants, five males and one female, aged between 22 and 26 years old (average age of 24.33). Regarding the dominant hand, four were right-handed (66.67%), one was left-handed (16.67%), and one indicated ambidexterity

(16.67%). The level of contact with the VR technology was gauged on a scale of one to five, and 16.67% of participants indicated a knowledge level of five, while the remaining 83.33% responded with a level of three. The preliminary tests were conducted primarily to refine the approach and prepare the methodology for the regular user study.

Regular study's phase one included 15 volunteers, six female and nine male, aged between 21 and 62 years old, with an average age of 28.07 (24.17 for females and 30.67 for males). The majority (86.67%) are right-handed, while 13.33% indicated ambidexterity (two male and two female). A few participants exhibited vision impairments (33.33% indicated a need for glasses, all male); however, these were not at a limiting level factor for the tasks that are a part of the study. Regarding familiarity with VR, 40% reported no prior exposure to the technology, 20% had been exposed to the device but had not attempted to use it, 26.67% had minimal experience with VR, and 13.33% had more extensive experience.

Two test modalities were conducted: one without a VR headset and one with it, both monitored for EEG and ECG signals. Non-VR tests offered a controlled, stable environment for data collection, while VR tests introduced an immersive, dynamic setting, adding complexity from increased sensory stimuli and movement. This comparative approach is essential for evaluating the real-time effects of VR immersion on brain and heart activity. Simultaneous EEG and ECG monitoring across both modalities offers a comprehensive view of physiological responses, enabling the analysis of brain activity patterns, heart rate variations, and movement-induced artifacts. Comparing these data provides valuable insights into VR's impact on therapeutic and biofeedback applications, particularly regarding signal quality and participant responses.

To facilitate the accurate examination of the behaviour exhibited by each participant, all sessions were recorded using a Nikon D5500 mounted on a tripod to monitor participants' movements. The room disposition included a blank wall without distractions in front of the participant. Data was collected using both technological and biological methods. A web platform [1] was employed to store and display patient performance across therapeutic sessions. The Pikita VR Quest SG [4], a specialized tool for food aversion therapy, was integrated with the platform. EEG signals were captured using the Unicorn Hybrid Black cap. This device has eight electrodes configured to capture the channels Fz, Cz, C3, C4, Pz, PO7, PO8, and Oz. EEG data were processed using EEGLAB. The ECG signal was captured using the BITalino. The EEG analysis is based on data derived from memory tasks and activities involving auditory stimuli. The Fz electrode is associated with attention and emotional regulation as it is located near the prefrontal cortex. Conversely, the C3 and C4 electrodes are linked to movement and sensory processing as they are located in the sensorimotor cortex. This study also aimed to identify the N200 and P300 paradigms in EEG signals recorded during each VR-based activity.

Participants performed the following tasks without VR: i) Baseline (remain with the eyes closed for five seconds, open the eyes, blink three times and repeat the process for one minute); ii) Relaxation (follow an audio of guided relaxation with the eyes closed for one minute); iii) Memorisation (read a list of ten words for

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one minute and repeat the list after a few seconds); iv) Observation and reaction (watch an animation video and grab an object when a crocodile appeared). On the other hand, the VR tasks include looking, reaching, grabbing, and simulating eating apples in the Pikita VR Quest SG (each action repeated three times). Finally, participants were asked to fill out a usability questionnaire concerning the tasks and devices used during the session.

3 USER STUDY - RESULTS

Setting thresholds to validate biomarker values (N200 required amplitudes below $-1 \mu V$, while P300 required values above $4.5 \mu V$), indicated that the N200 was the most consistently detected biomarker across participants, while P300 only occasionally approached the threshold. The findings suggest that N200 is more reliably detected in VR tasks, while P300 may be harder to capture, possibly due to task characteristics. This highlights the potential of EEG biomarkers for assessing cognitive engagement in VR, pointing to the need for refined measurement techniques.

The memorization task (without VR) aimed to assess the occurrence of significant brain activation during cognitively demanding activities, specifically during a word-list memorization task and, subsequently, during the task of recalling the memorized words (Figure 1B). The results indicate a significant increase in brain activation during both tasks compared to baseline. Memorization showed heightened beta frequency activity, reflecting enhanced attention and mental focus. Similarly, recall tasks demonstrated increased activation in the beta and higher frequency bands, signaling the neural effort required to access and organize stored information. These findings highlight the brain's dynamic adjustment to the specific demands of each task phase.

Independent Components (ICs) were extracted by performing Independent Component Analysis on the EEG data acquired with and without VR. The time-frequency analysis on component IC3 revealed activity in the beta frequency band. The presence of beta band activity is particularly relevant, as this range is frequently associated with cognitive processes such as attention, focus, and working memory, making it the target frequency for the task at hand. This result in IC3 confirms that the captured signal reflects the expected brain activity, indicating that neural processing in the beta frequency band is directly involved in task performance. This measure is particularly useful in studies on attention and memory as it captures the dynamic changes in neural processing associated with external stimuli.

Event-related potentials (ERP) analysis was conducted for each experimental activity (memorization and word repetition), and the average baseline value was subtracted from the ERP values obtained for each activity, isolating the specific amplitude changes related to cognitive activities, and providing a more precise measure of the desired biomarkers. This procedure enables not only the detection of changes in neural response times but also the establishment of consistent parameters for NF use. Since ERPs offer insight into shifts in mental state and attention, acquiring reliable N200 and P300 values from the Fz channel is of importance for NF applications. It provides insights into the brain's responsiveness to task stimuli, which can be utilized to enhance interventions to optimize memory and focus through brain training. The N200 values for Participant 9 ($-3.67 \mu V$) and the P300 values for Participant 15 ($4.65 \mu V$) fall within the expected ranges for their respective biomarkers. This outcome, though more of a tendency, supports the validity of using these biomarkers.

The grabbing and eating task (with VR) aims to assess whether cerebral habituation would occur (Figure 1A). Participants were asked to perform a task of grabbing and eating three apples in sequence, repeating the same action three times. The methodology used consists of precise event marking (at every grasp) and EEG signal segmentation (three separate analyzes using a time window



Figure 1: Experimental set-up A) Test with VR. B) Test without VR.

of $20 \mu s$ before and after the event). By comparing the EEG signals across the three apple-grasping events, patterns of neural activity were examined to identify any reductions in response intensity or changes in activation patterns that might indicate habituation. The analysis showed that, in 12 cases, there was a progressive decrease in EEG signal amplitude over the three instances of the action, indicating habituation. These findings suggest that with repetition, the brain requires fewer resources to perform the same motor task, demonstrating neural adaptation.

Although ECG data were collected during the exercises, the analysis of these data was not extensive because of noise and artifacts in the ECG data due to the experimental context where participant movement played a significant role. The ECG signal presented good quality when participants were still.

4 CONCLUSION AND FUTURE WORK

The study confirmed that EEG reliably captured task-dependent brain activity, and the concurrent use of EEG, ECG, and VR posed no impediments for participants, thus establishing a valuable foundation. Future work will focus on refining the system, expanding testing to pediatric populations, particularly children with food aversions, and enhancing user comfort through adaptation.

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